

By examining how suppression and amplification effects change across the model’s depth, we discovered a clear pattern. Early layers (0–7) tend to suppress forbidden concepts, consistent with the model initially trying to follow instructions. Middle layers (8–16) show the most variability, with both strong suppression and rebound effects. Later layers (20+) generally have smaller effects, suggesting the major decisions about forbidden tokens are made earlier in processing. Figure 3 illustrates this progression, showing both the average effect per layer and the count of suppressive versus amplifying heads. The pattern reveals a kind of internal struggle; the model initially attempts suppression but then partially reverses course through amplifying heads in middle layers.

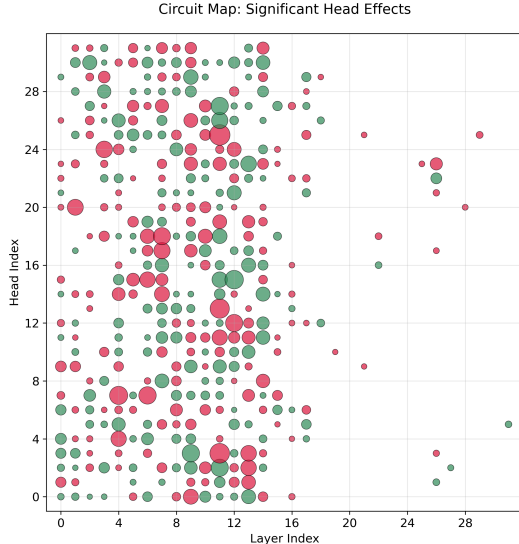


Figure 4: Circuit map of the most influential attention heads. Circle size indicates effect magnitude; color indicates direction (red = amplification, green = suppression). Effects cluster in the middle layers.

Despite involving over 1,000 attention heads, the circuits responsible for ironic rebound are remarkably sparse. Typically, just 15–20 heads (roughly 1–2 per layer, concentrated in middle layers) account for 80% or more of the total rebound effect. Figure 4 visualizes these key heads, showing their positions and effect magnitudes. The strongest suppressors tend to appear around layers 8–13, while the most potent amplifiers concentrate in layers 12–17.

This localization is practically important; it suggests that ironic rebound could potentially be mitigated by targeted interventions on specific heads, rather than requiring holistic model changes. Across the full model architecture, 26.7% of heads contribute to rebound, 29.7% to suppression, and 43.7% remain functionally neutral. This suggests that more robust negation following might be achievable through precision modifications rather than holistic architectural changes. See Appendix C for further results.

4 Conclusion

LLMs display ironic rebound: after a “do not mention X” instruction, they become more likely to output X, with circuit tracing showing a sparse set of middle-layer attention heads that amplify the forbidden token under load even as early layers suppress it. Semantic distractors induce the strongest rebound, while simple repetition suppresses the most. Models with sharper polarity separation show longer-lasting rebound, reflected in higher L50 values. Despite limits such as a synthetic English-only dataset, log-probability evaluations, and a small set of distractor types, the results expose the fragility of negation-based filters and explain why this fragility can persist across context lengths, since sparse amplification heads can overpower early suppression when load and framing align, motivating tests in natural dialogue and with multi-word targets.

References

- [1] Sandhini Agarwal, Lama Ahmad, Jason Ai, Sam Altman, Andy Applebaum, Edwin Arbus, Rahul K Arora, Yu Bai, Bowen Baker, Haiming Bao, et al. gpt-oss-120b & gpt-oss-20b model card. *arXiv preprint arXiv:2508.10925*, 2025.
- [2] Ameen Ali, Lior Wolf, and Ivan Titov. Mitigating copy bias in in-context learning through neuron pruning. *arXiv preprint arXiv:2410.01288*, 2024.
- [3] Miriam Anschutz, Diego Miguel Lozano, and Georg Groh. This is not correct! negation-aware evaluation of language generation systems, 2023.
- [4] Stella Biderman, Hailey Schoelkopf, Quentin Anthony, Herbie Bradley, Kyle O’Brien, Eric Hallahan, Mohammad Aflah Khan, Shivanshu Purohit, USVSN Sai Prashanth, Edward Raff, Aviya Skowron, Lintang Sutawika, and Oskar van der Wal. Pythia: A suite for analyzing large language models across training and scaling, 2023.
- [5] Sid Black, Stella Biderman, Eric Hallahan, Quentin Anthony, Leo Gao, Laurence Golding, Horace He, Connor Leahy, Kyle McDonell, Jason Phang, Michael Pieler, USVSN Sai Prashanth, Shivanshu Purohit, Laria Reynolds, Jonathan Tow, Ben Wang, and Samuel Weinbach. Gpt-neox-20b: An open-source autoregressive language model, 2022.
- [6] Aaron Grattafiori, Abhimanyu Dubey, Abhinav Jauhri, Abhinav Pandey, Abhishek Kadian, Ahmad Al-Dahle, Aiesha Letman, Akhil Mathur, Alan Schelten, Alex Vaughan, et al. The llama 3 herd of models. *arXiv preprint arXiv:2407.21783*, 2024.
- [7] Zhiyong Han, Fortunato Battaglia, Kush Mansuria, Yoav Heyman, and Stanley R. Terlecky. Beyond text generation: Assessing large language models’ ability to reason logically and follow strict rules. *AI*, 6(1), 2025.
- [8] Kyomin Hwang, Suyoung Kim, Junhoo Lee, and Nojun Kwak. Do not think about pink elephant! In *Proceedings of the IEEE/CVF Conference on Computer Vision and Pattern Recognition*, pages 8296–8303, 2024.
- [9] Liquid AI. Lfm2-350m. <https://huggingface.co/LiquidAI/LFM2-350M>, 2025. Liquid AI’s hybrid on-device model (350M parameters), featuring multiplicative gates and short convolutions and optimized for edge applications (fast training, fast inference, multilingual support).
- [10] Nelson F Liu, Kevin Lin, John Hewitt, Ashwin Paranjape, Michele Bevilacqua, Fabio Petroni, and Percy Liang. Lost in the middle: How language models use long contexts. *arXiv preprint arXiv:2307.03172*, 2023.
- [11] Callum McDougall, Arthur Conmy, Cody Rushing, Thomas McGrath, and Neel Nanda. Copy suppression: Comprehensively understanding an attention head, 2023.
- [12] Callum McDougall, Arthur Conmy, Cody Rushing, Thomas McGrath, and Neel Nanda. Copy suppression: Comprehensively understanding a motif in language model attention heads. In *Proceedings of the 7th BlackboxNLP Workshop: Analyzing and Interpreting Neural Networks for NLP*, pages 337–363, 2024.
- [13] Anca M Miron and Jack W Brehm. Reactance theory-40 years later. *Zeitschrift für Sozialpsychologie*, 37(1):9–18, 2006.
- [14] Alec Radford, Jeff Wu, Rewon Child, David Luan, Dario Amodei, and Ilya Sutskever. Language models are unsupervised multitask learners. In *NeurIPS Workshop on Deep Learning*, 2019.
- [15] Benjamin D Rosenberg and Jason T Siegel. A 50-year review of psychological reactance theory: Do not read this article. *Motivation Science*, 4(4):281, 2018.
- [16] Elsa Skänderi-Rakiplari. Semantic priming effects of synonyms, antonyms, frame, implication and verb-object categories. *Logos & Littera: Journal of Interdisciplinary Approaches to Text*, 2(4), 2017.

- [17] Yeonkyoung So, Gyuseong Lee, Sungmok Jung, Joonhak Lee, JiA Kang, Sangho Kim, and Jaejin Lee. Thunder-nubench: A benchmark for llms’ sentence-level negation understanding, 2025.
- [18] Christina Steindl, Eva Jonas, Sandra Sittenthaler, Eva Traut-Mattausch, and Jeff Greenberg. Understanding psychological reactance. *Zeitschrift für Psychologie*, 2015.
- [19] Neeraj Varshney, Satyam Raj, Venkatesh Mishra, Agneet Chatterjee, Ritika Sarkar, Amir Saeidi, and Chitta Baral. Investigating and addressing hallucinations of llms in tasks involving negation. *arXiv preprint arXiv:2406.05494*, 2024.
- [20] Tereza Vrabcová, Marek Kadlčík, Petr Sojka, Michal Štefánik, and Michal Spiegel. Negation: A pink elephant in the large language models’ room? *arXiv preprint arXiv:2503.22395*, 2025.
- [21] Deming Wang, Martin S Hagger, and Nikos LD Chatzisarantis. Ironic effects of thought suppression: A meta-analysis. *Perspectives on Psychological Science*, 15(3):778–793, 2020.
- [22] Daniel M Wegner, Ralph Erber, and Sophia Zanakos. Ironic processes in the mental control of mood and mood-related thought. *Journal of personality and social psychology*, 65(6):1093, 1993.
- [23] Daniel M Wegner, David J Schneider, Samuel R Carter, and Teri L White. Paradoxical effects of thought suppression. *Journal of personality and social psychology*, 53(1):5, 1987.
- [24] BigScience Workshop, Teven Le Scao, Angela Fan, Christopher Akiki, Ellie Pavlick, Suzana Ilić, Daniel Hesslow, Roman Castagné, Alexandra Sasha Luccioni, François Yvon, et al. Bloom: A 176b-parameter open-access multilingual language model. *arXiv preprint arXiv:2211.05100*, 2022.
- [25] An Yang, Anfeng Li, Baosong Yang, Beichen Zhang, Binyuan Hui, Bo Zheng, Bowen Yu, Chang Gao, Chengen Huang, Chenxu Lv, et al. Qwen3 technical report. *arXiv preprint arXiv:2505.09388*, 2025.
- [26] Susan Zhang, Stephen Roller, Naman Goyal, Mikel Artetxe, Moya Chen, Shuohui Chen, Christopher Dewan, Mona Diab, Xian Li, Xi Victoria Lin, et al. Opt: Open pre-trained transformer language models. *arXiv preprint arXiv:2205.01068*, 2022.

A Appendix

A.1 Dataset Details

Sources

ReboundBench combines and extends two public negation datasets:

- **This-is-not-a-dataset** (HiTZ)[3]
- **NUBench** (SNU Thunder)[17]

Preparation

We extracted English examples, selected common nouns that tokenize as a single token, and normalized text (lowercasing, removing punctuation). Each item was wrapped in a template of the form:

```
[Topic]. Do not mention [Target]. [Distractor text] ...
```

Distractors varied by *type* (semantic, syntactic, repetition) and *length* (0–1024 tokens). For polarity discrimination, each (topic, target) pair was expanded into negative/neutral continuations. The final dataset contains about 5,000 items in CSV format with fields: topic, target, distractor type, distractor length, and prompt.

Examples

- Topic: Paris travel; Target: `eiffel`; Repetition distractor: Write about a trip to Paris. Do not mention the Eiffel. Paris is a place many tourists visit.”
- Topic: Farm animals; Target: `sheep`; Semantic distractor: Write about animals on a farm. Do not mention sheep. Cows and pigs are often raised for food.”

Release

We provide the dataset and scripts for generation as a CSV file with $\sim 5k$ prompts for reproducibility and further study.